UNIVERSITÀ
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CONTRASTIVE LEARNING FOR ROBUST AUTONOMY IN UAVS

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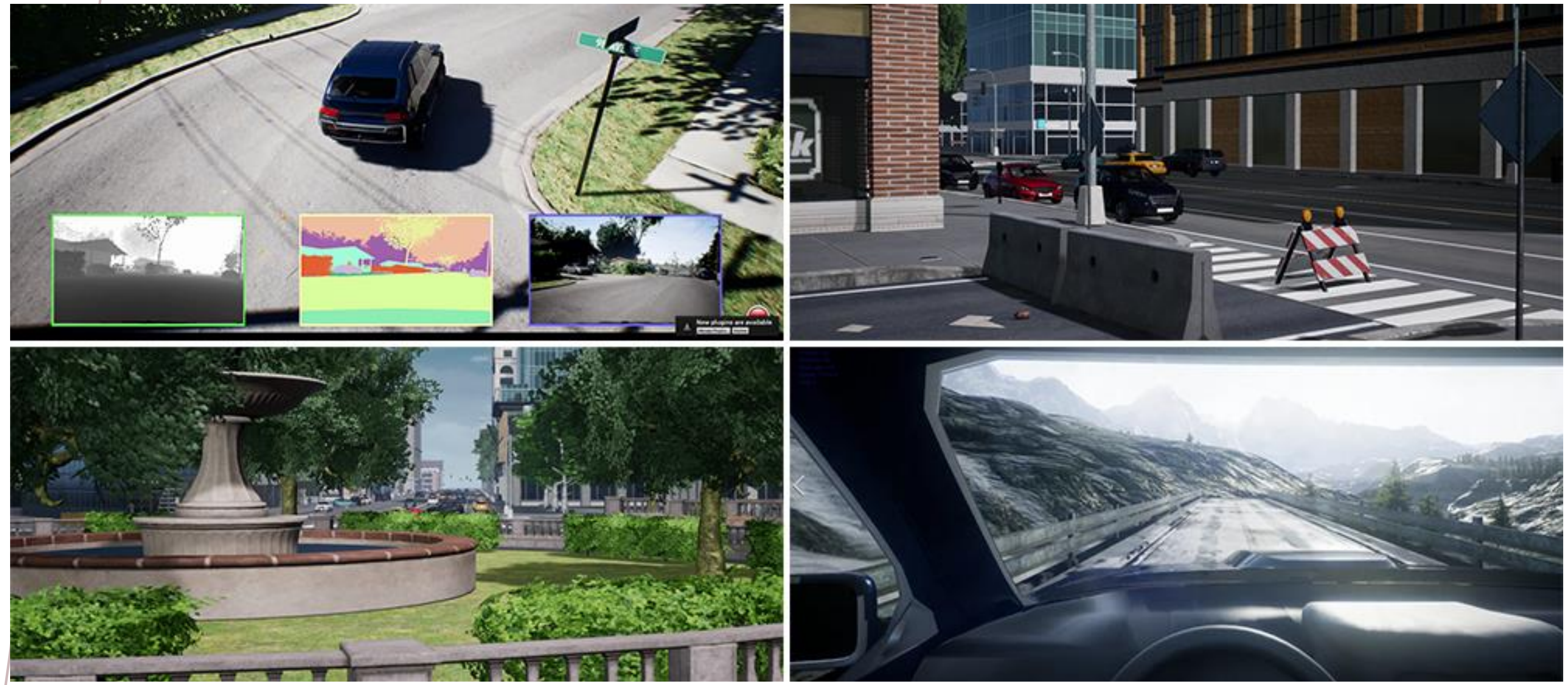
ABSTRACT

This project investigates the use of contrastive learning to enhance the robustness of flight control in UAVs. To this end, a dataset of images was generated through AirSim simulations (an open-source simulator developed by Microsoft that provides realistic environments for training and validating autonomous vehicle control systems, such as drones and cars), covering different environments and flight conditions. Then the dataset is used to train a vision encoder that maps images into an embedding space, learning to categorize them and distinguish obstacles from other objects.

DATASET

AIRSIM



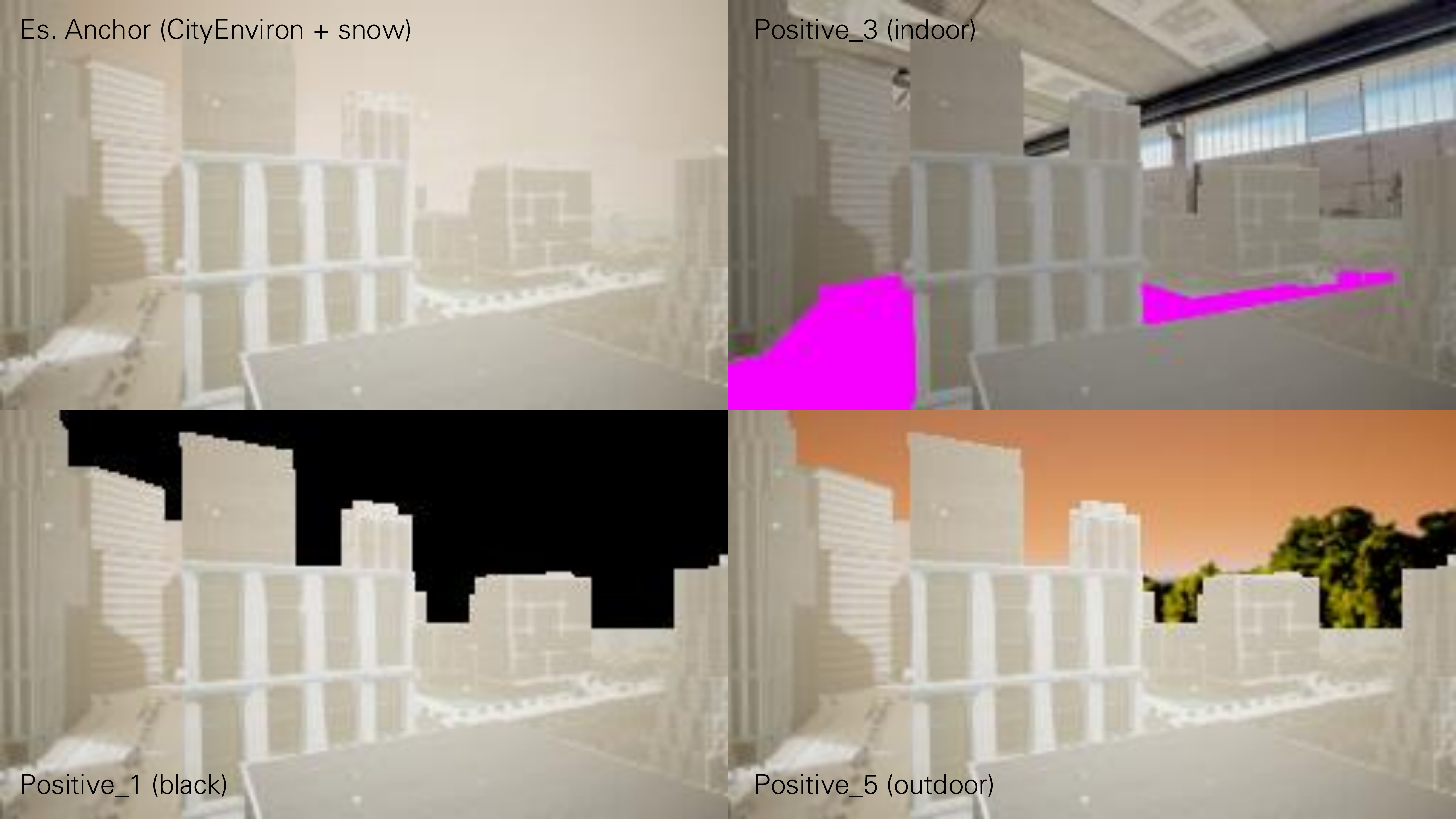
AirSim is an open-source simulator for drones, cars and more, built on *Unreal Engine*.

It was developed as a platform for AI research to experiment with deep learning, computer vision and reinforcement learning algorithms for autonomous vehicles. For this purpose, AirSim also exposes APIs to retrieve data and control vehicles in a platform independent way.

DATASET STRUCTURE

In order to the CL trainer, the dataset is organized on anchors and their positive pairs (background replacement), in differents environments:

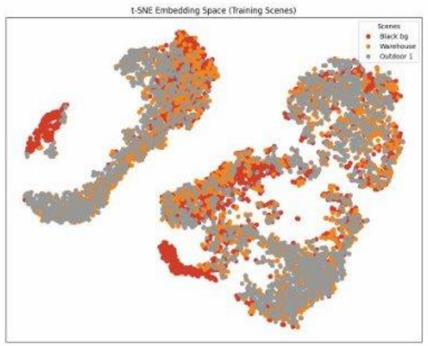
- *12500 anchors, with 6 positive pairs for each one:*
  - *Black and white background replcacement*
  - *One indoor positive, with magenta bottom ground replacement*
  - *Three outdoor positive*
- *Blocks of 2500 anchors for every environment (Neightborhood, City, Coastline, AbandonedPark)*
  - *Weather variations with rain, fog, snow, sun envs for major dataset variety .*
- *The negatives are all the others anchors*

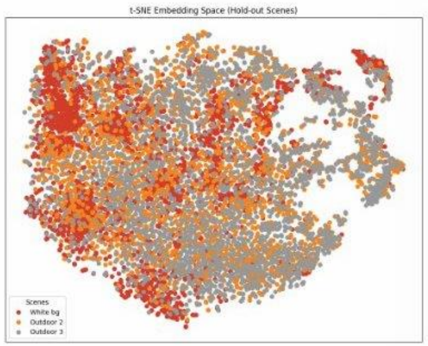

Es. Anchor (CityEnviron + snow)
Positive_3 (indoor)
Positive_1 (black)
Positive_5 (outdoor)

TRAINING

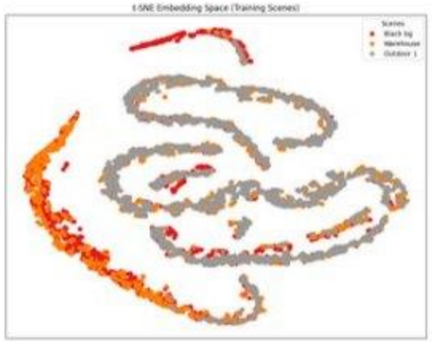
Objectives:

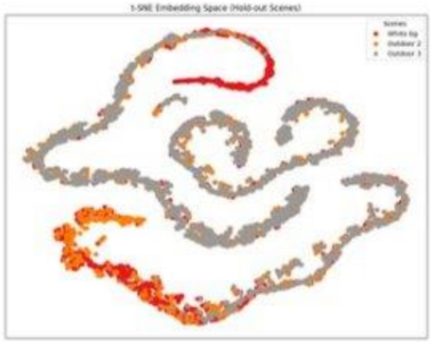
1. High Hold-out consistency
2. Low loss
3. Good-quality embedding space:



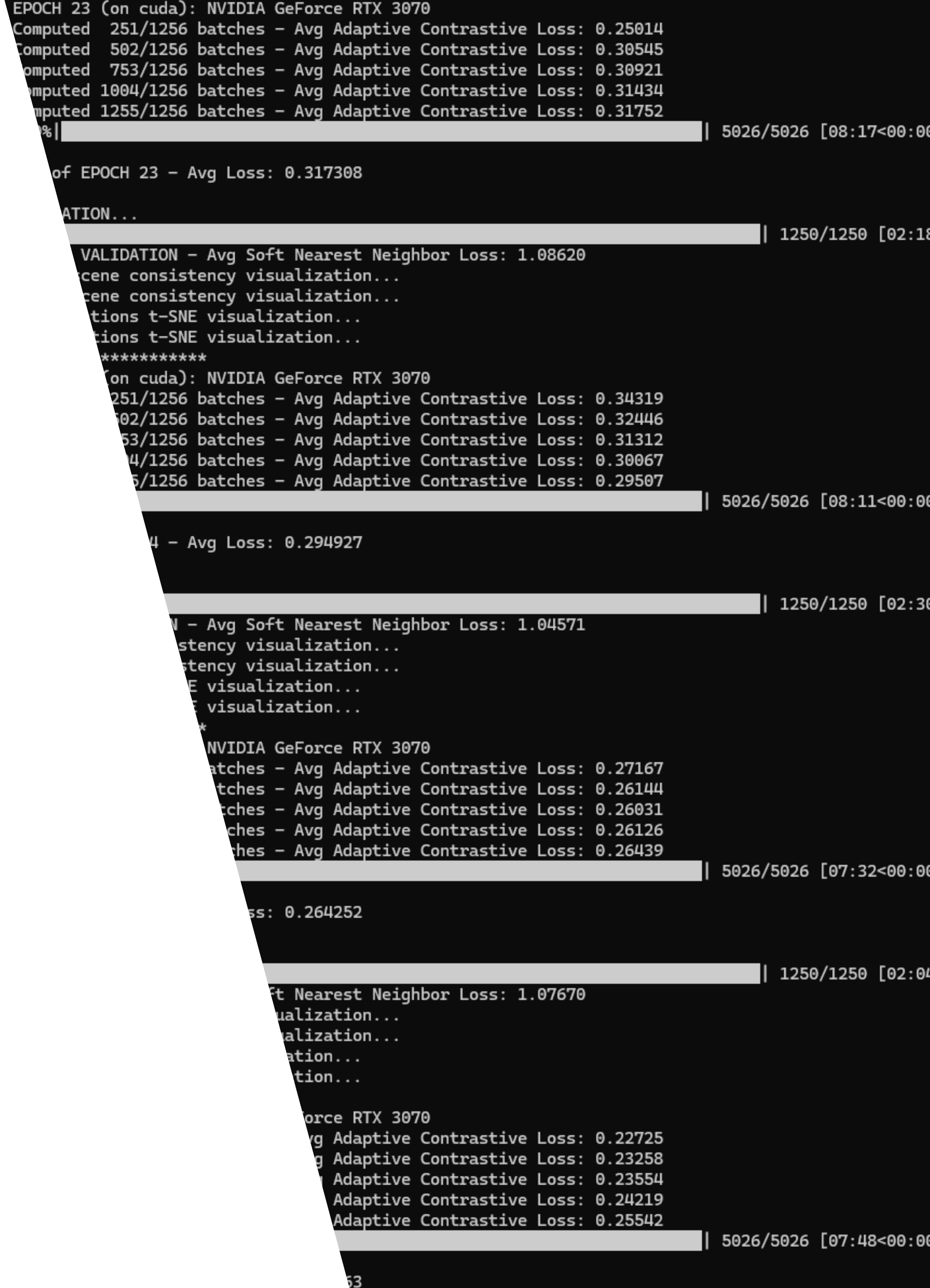


Bad-formed clusters





Well-formed clusters



CONTRASTIVE LEARNING

CL learns representations by comparing examples: similar pairs are pulled together, dissimilar pairs pushed apart in the latent space.

Data does not need to be labeled, using augmentations to create positives and other anchors as negatives.

In my dataset, positive pairs are automatically generated, separating obstacles from ground and background, and replacing them.

However, for negative pairs, since some anchors are very similar to each other, it is necessary to adapt the temperature in order to improve training.

LOSS WITH DYNAMIC TEMPERATURE

The **loss value** drives the learning in contrastive frameworks by pulling similar examples together and pushing dissimilar examples apart in the latent space.

The temperature scales the loss into the training:

$$L_i = -log \frac{\exp\left(\frac{sim(z_i\,,\ z_i^+)}{\tau}\right)}{\sum_{k \neq i} \exp\left(\frac{sim(z_i\,,\ z_k)}{\tau}\right)}$$

Since many anchors are similar to each other, it was necessary to define a function that, starting from **privileged data** (position, angle) for each anchor, calculates the expected similarity in a pair of anchors.

$$D_{pos} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

$$S_{pos} = e^{-D_{pos}}$$

$$S_{rot} = \left|Q_1 * Q_2\right|$$

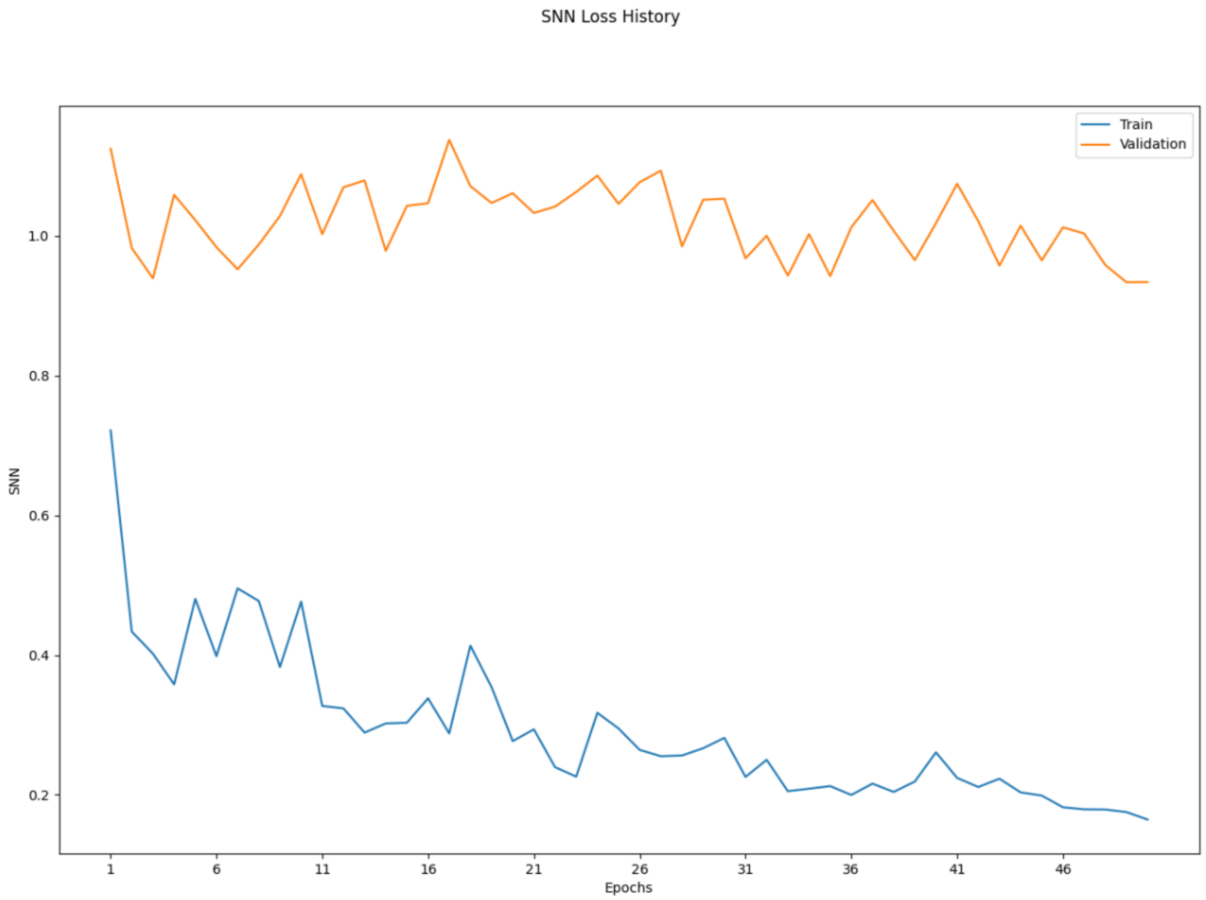
$$SA = \left(S_{pos} * w_{pos}\right) + \left(S_{rot} * w_{rot}\right)$$

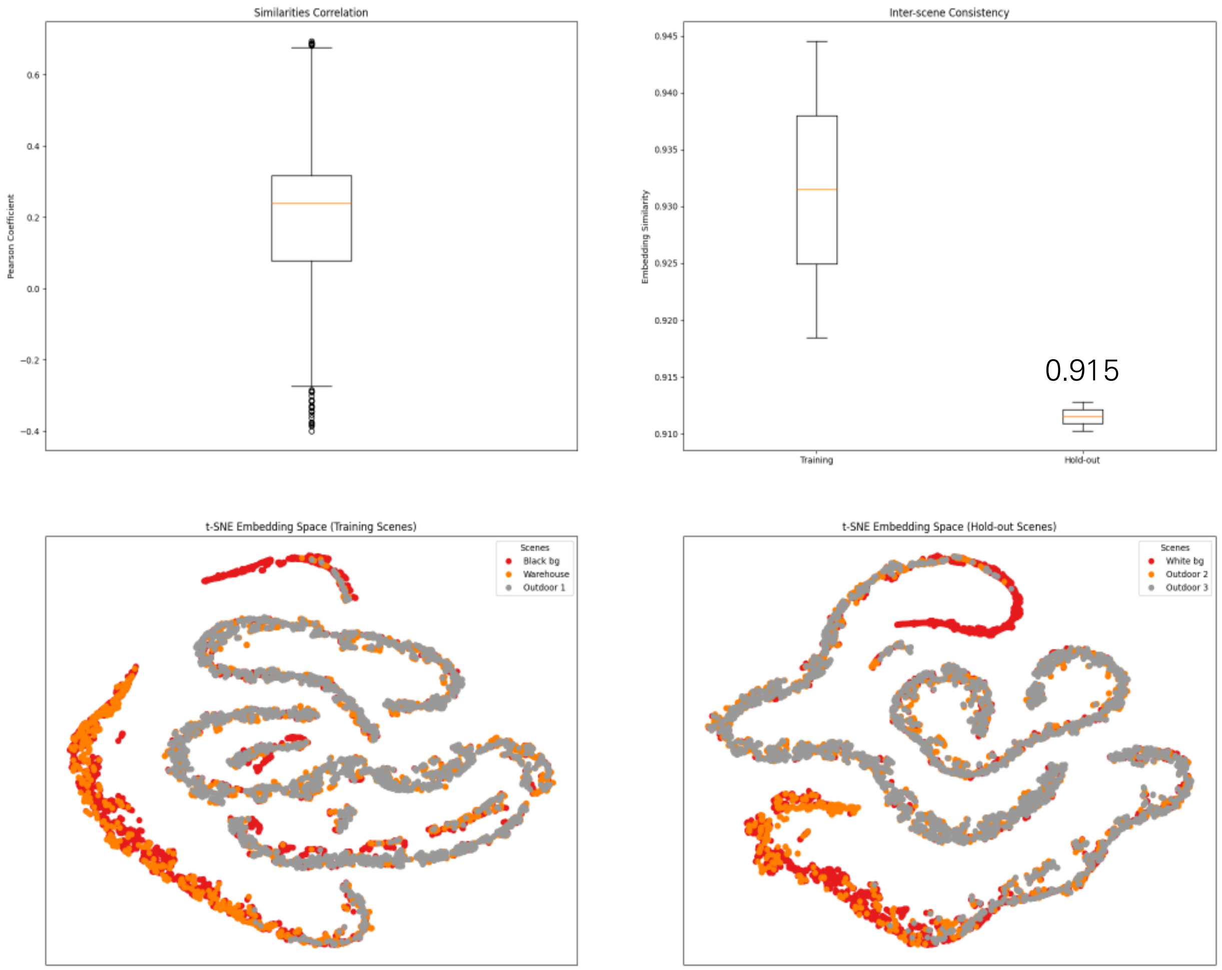
$(x, y, z) \rightarrow vector\ position$

$Q \rightarrow rotation\ (quaternion)$

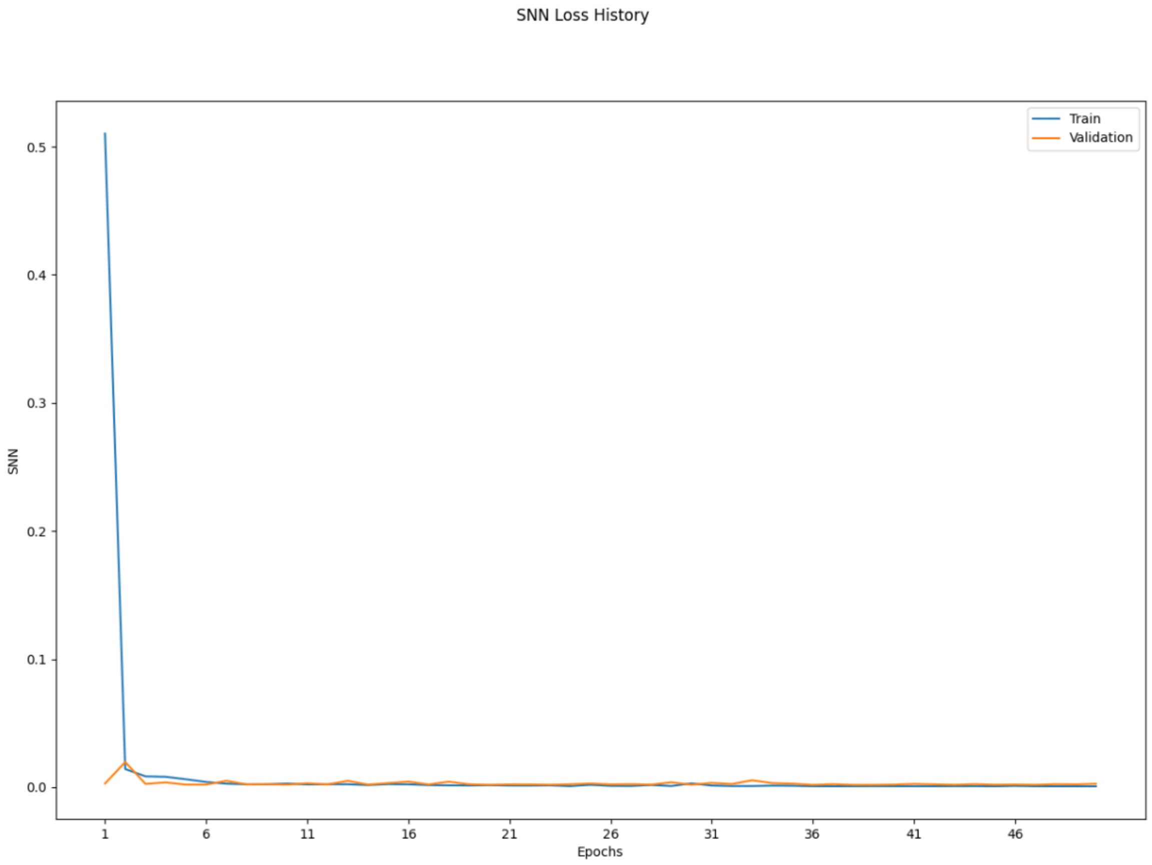
$w_{pos}\,, w_{rot} \rightarrow weights \in [0,1]$

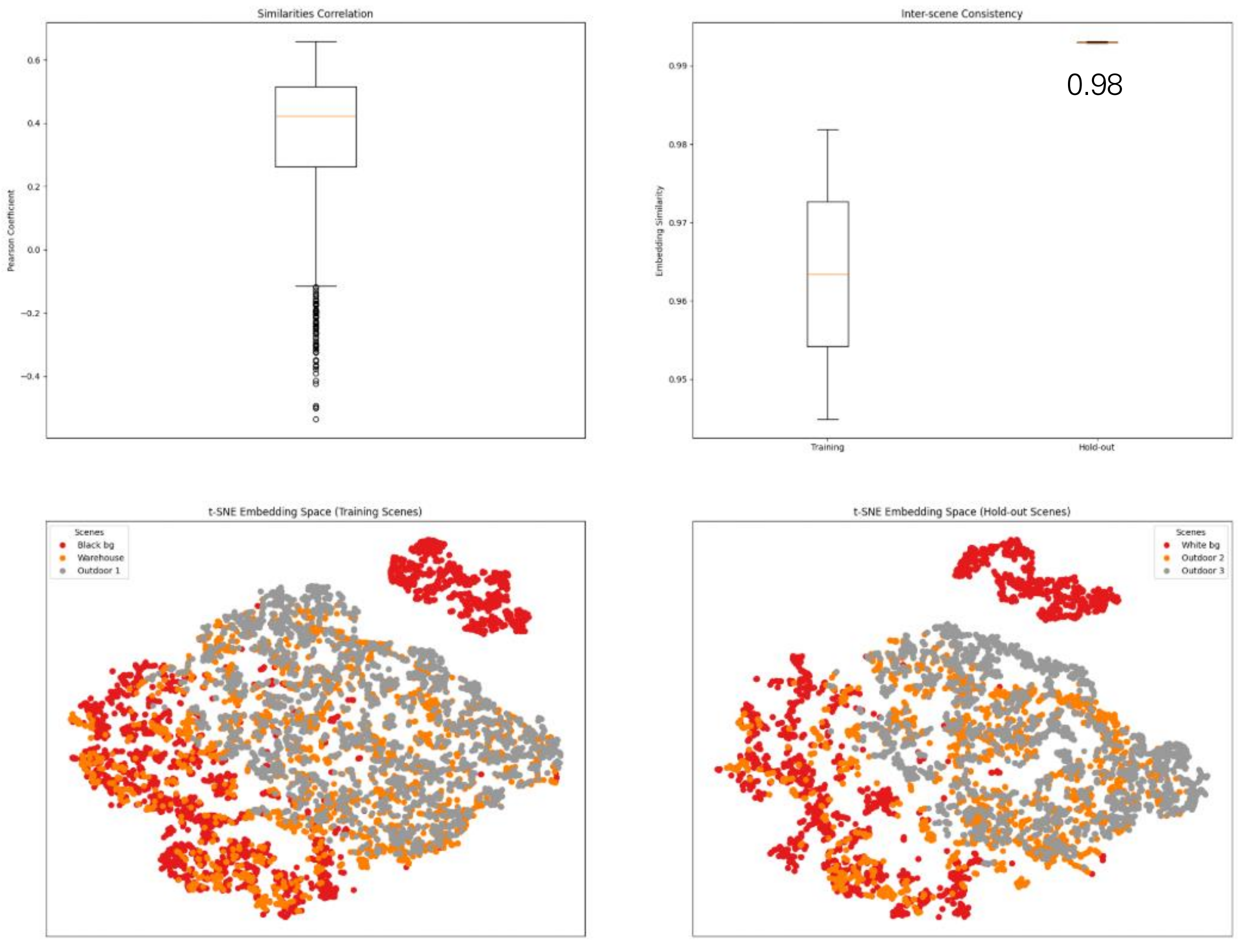
STATIC TEMPERATURE





DYNAMIC TEMPERATURE





TRAINING RESULT

The training followed two phases: the first one was about find out how the changing parameters affects results, and understand why.

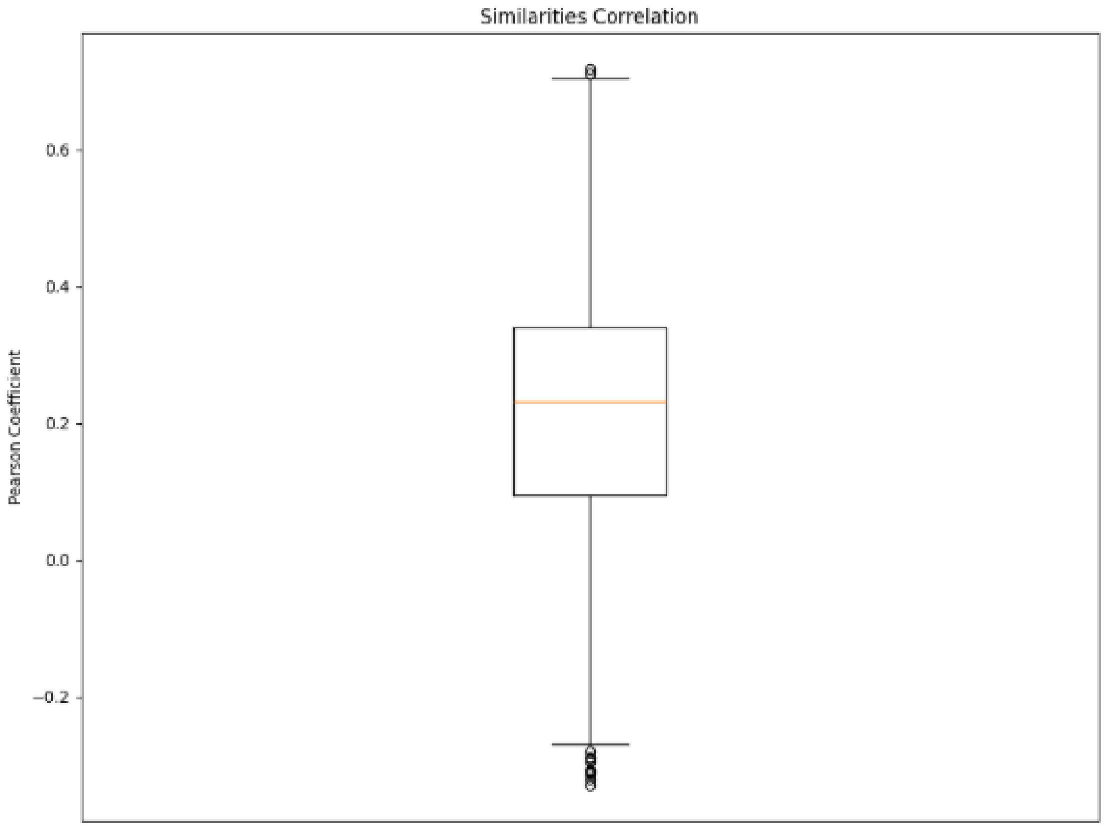
Then, the 2nd phase, was about trying to reach out one of the better possible configuration.

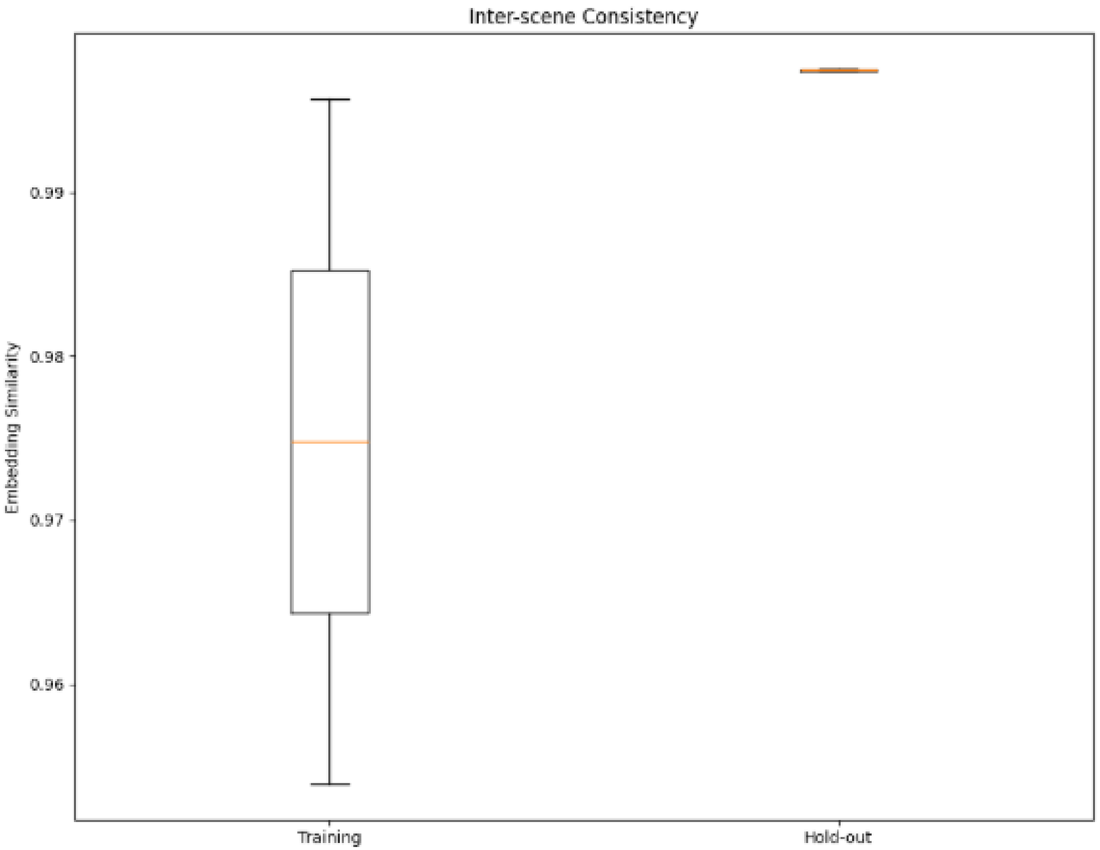
The model used is resnet50, on the *simclr* framework.

Experiment	ES quality	Loss	Hold-out Consistency
Static temperature Static on 0.7 Batch=8, mb=2	optimal	0.175278	0.891
+ Dinamic temperature Min=0.1, Max=1	not optimal	0.000707	0.987
+ Higher batch Batch=2048, mb=16	better	0.000700	0.994
Final configuration Batch=2048, mb=16 Tau: min=0.4, max=5 Learning rate = 0.0003	good	0.055835	0.999

FINAL CONFIGURATION

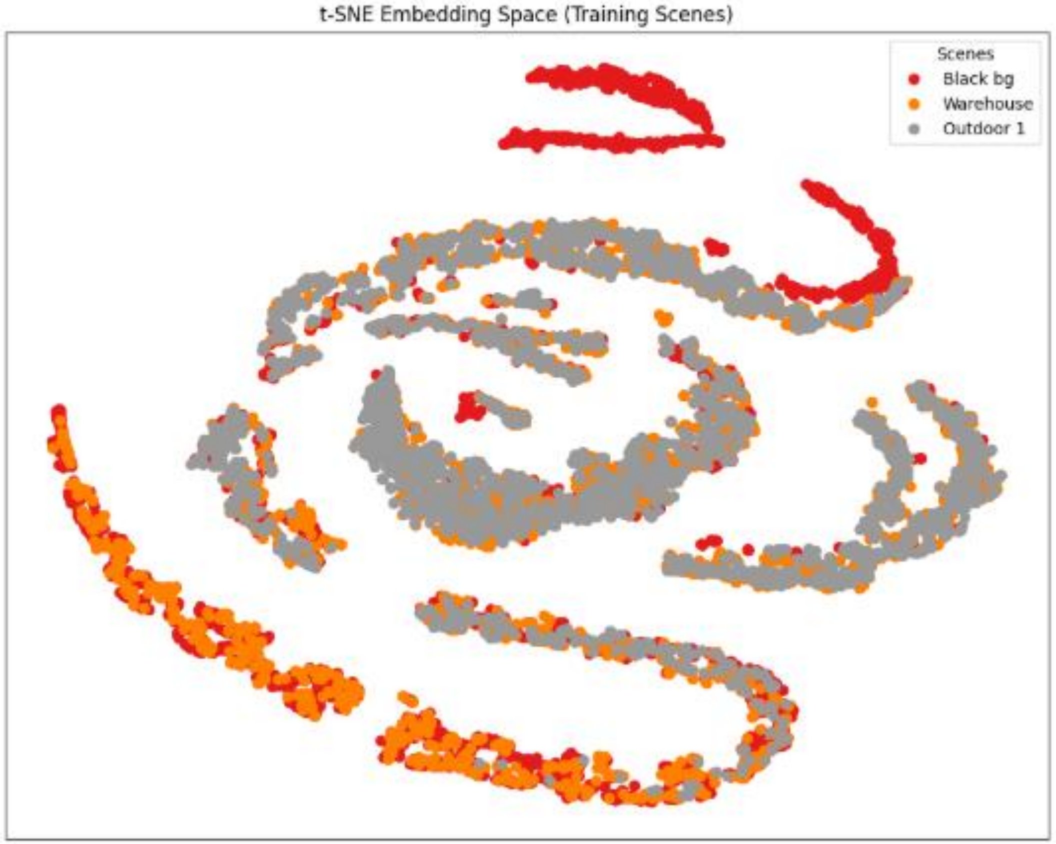
Similarity - Consistency

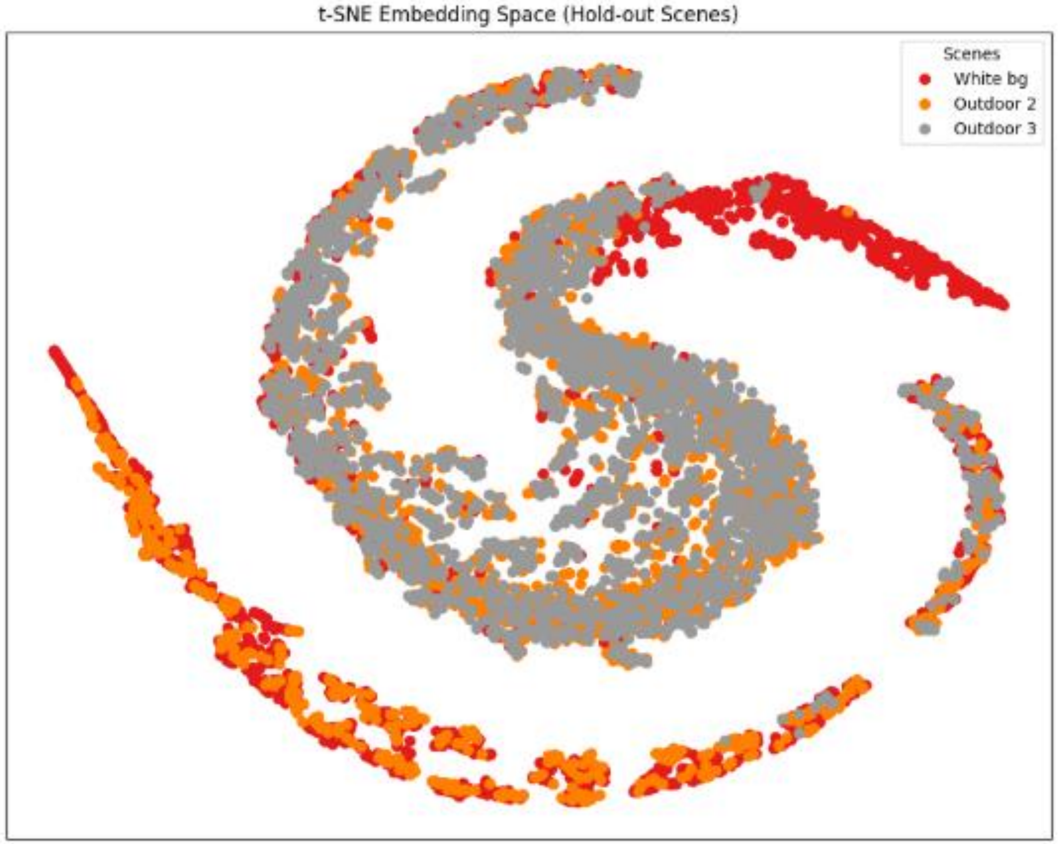




FINAL CONFIGURATION

Embedding Space





FINAL REMARKS

With the final configuration, I've reached a robust vision encoder,
but it is not enough to control the drone.
For this, it would be necessary to develop an entire control policy through reinforcement learning, based on the model developed in this work.

GitHub project link

Thank you for your attention.